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| **Name of Course** | **Bachelor of Computer Application (BCA)** |
| **Semester** | **VI** |
| **Name of Subject** | **Cyber Security** |
| **Subject Code** | **BCA-605-B** |
| **Marks** | **75 Marks** |
| **Lectures** | **50 Lectures** |

**Unit I**

**1. IT Act and Encryption**

**1.1 Object of the Act**
**1.2 Scope of the Act**
**1.3 Symmetric Cryptography**
**1.4 Asymmetric Cryptography**
**1.5 RSA Algorithm**
**1.6 Public Key Encryption**

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## 1.1 Object of the Act:

The **object of the Act** in cyber security—mainly referring to laws like the **Information Technology (IT) Act, 2000** and its amendments—is to provide a **legal framework** for regulating activities in the digital environment. Its primary objective is to ensure **safe, secure, and reliable use of computers, networks, and electronic transactions**.

The key objects include:

1. **Legal Recognition of Electronic Records and Digital Signatures:**
   The Act aims to promote electronic communication and e-commerce by giving legal validity to digital documents and electronic signatures.

2. **Protection Against Cybercrimes:**
   It defines various cyber offences such as hacking, identity theft, cyberstalking, and unauthorized access, and prescribes penalties to deter such activities.

3. **Safeguarding Privacy and Data Security:**
   The Act sets guidelines for protecting sensitive personal data and ensuring confidentiality, integrity, and availability of information systems.

4. **Regulation of Intermediaries and Cyber Entities:**
   It outlines responsibilities of service providers, social media platforms, and intermediaries to prevent misuse of their platforms and to cooperate with law enforcement agencies.

5. **Promoting Trust in Digital Transactions:**
   By enforcing cyber laws and securing online systems, the Act builds user confidence in e-governance, e-banking, and digital services.

**In short:**

The object of the Act in cyber security is **to create a secure digital ecosystem, prevent**

**cybercrimes, protect data, regulate online activities, and promote safe electronic transactions**.

## 1.2 Scope of the Act:

The **scope of the Information Technology (IT) Act, 2000** defines the range of activities, sectors, and digital operations that the law covers. It applies to all electronic records, digital transactions, and cyber activities occurring within India or affecting Indian systems. The main points of its scope are:

1. **Applicability to Electronic Records and Digital Signatures:**
   The Act governs the creation, storage, transmission, and authentication of electronic records and digital signatures, giving them legal recognition.

2. **Coverage of Cybercrimes and Offences:**
   It includes offences such as hacking, data theft, cyberstalking, identity theft, phishing, and unauthorized access. It applies to individuals, companies, and organizations.

3. **Regulation of Intermediaries and Cyber Service Providers:**
   The Act covers ISPs, social media platforms, e-commerce companies, and network service providers, defining their obligations and liability.

4. **Jurisdiction Beyond Physical Boundaries:**
   The Act applies to cyber offences committed **within India or outside India** if the offence involves a computer or network located in India.

5. **E-Governance and Digital Transactions:**
   It covers electronic contracts, e-filing, e-payments, and digital communication with government agencies, promoting secure and efficient e-governance.

**In short:**

The scope of the Act extends to **all electronic records, digital transactions, cyber offences, intermediaries, and computer-related activities**, even if performed outside India but impacting Indian systems.

## 1.3 Symmetric Cryptography:

**Symmetric cryptography**, also called **secret-key cryptography**, is an encryption technique where the **same key** is used for both **encryption and decryption** of data. It is one of the oldest and fastest methods of securing information.

**Key Features:**

1. **Single Key Usage:**
   A common secret key is shared between the sender and receiver. The security of the system depends on keeping this key confidential.

2. **High Speed and Efficiency:**
   Symmetric algorithms are computationally faster, making them suitable for encrypting large amounts of data such as files, databases, and communication streams.

3. **Common Algorithms:**
   Popular symmetric algorithms include **AES (Advanced Encryption Standard)**, **DES**

**(Data Encryption Standard)**, and **Blowfish**.

4. **Applications:**
   Used in secure communication, data storage, VPNs, and wireless security. It is widely used in real-time systems requiring quick data processing.

5. **Limitation – Key Distribution Problem:**
   The major challenge is securely sharing the secret key between sender and receiver. If the key is exposed, the entire communication becomes insecure.

**In short:**

Symmetric cryptography is a fast and efficient method of encryption that uses a **single shared secret key**, but faces challenges in **key distribution**.

## 1.4 Asymmetric Cryptography:

**Asymmetric cryptography**, also known as **public-key cryptography**, is an encryption technique that uses **two different keys**: a **public key** for encryption and a **private key** for decryption. These keys are mathematically related but cannot be derived from each other easily, which ensures strong security.

**Key Features:**

1. **Two-Key System:**
   Each user has a **public key** (shared with everyone) and a **private key** (kept secret). Data encrypted with one key can only be decrypted using the other.

2. **High Security for Key Exchange:**
   It solves the **key distribution problem** of symmetric cryptography. The sender does not need to share a secret key; they can use the receiver's public key.

3. **Digital Signatures:**
   Asymmetric cryptography enables authentication and non-repudiation using digital signatures. A message signed with a private key can be verified using the corresponding public key.

4. **Common Algorithms:**
   Popular algorithms include **RSA (Rivest–Shamir–Adleman)**, **DSA (Digital Signature Algorithm)**, and **Elliptic Curve Cryptography (ECC)**.

5. **Applications:**
   Used in secure email (PGP), SSL/TLS for website security, digital certificates, blockchain, and secure key exchange in many protocols.

**In short:**

Asymmetric cryptography uses a **public key–private key pair** to provide secure communication, authentication, and digital signatures, solving the limitations of symmetric key exchange.

## 1.5 RSA Algorithm:

**RSA (Rivest–Shamir–Adleman)** is one of the most widely used **asymmetric cryptographic algorithms**. It uses a **public key** for encryption and a **private key** for decryption. The security of RSA is based

on the mathematical difficulty of **factorizing large prime numbers**.

**Steps of the RSA Algorithm:**

1. **Key Generation:**

   Select two large prime numbers **p** and **q**.
   Compute **n = p × q**, which becomes part of the public and private key.
   Calculate **φ(n) = (p – 1)(q – 1)**.
   Choose a public key **e** such that it is relatively prime to φ(n).
   Compute the private key **d**, where **d × e ≡ 1 (mod φ(n))**.

2. **Encryption:**

   The sender uses the **receiver's public key (e, n)**.
   Ciphertext is computed as:
   $\mathbf{C = M^e \bmod n}$,
   where **M** is the plaintext message.

3. **Decryption:**

   The receiver uses their **private key (d, n)**.
   Plaintext is recovered as:
   $\mathbf{M = C^d \bmod n}$.

4. **Security:**

   RSA is secure because factoring **n** into its prime components (p and q) is extremely difficult for large key sizes (1024–4096 bits).

5. **Applications:**

   Used in **digital signatures**, **SSL/TLS**, **email encryption (PGP)**, and **secure key exchange**.

**In short:**

The RSA algorithm is a **public-key cryptographic scheme** that provides secure encryption and digital signatures by using **two keys** and the complexity of factoring large primes.

## 1.6 Public Key Encryption:

**Public Key Encryption**, also known as **asymmetric encryption**, is a cryptographic technique where two different keys are used for secure communication: a **public key** for encryption and a **private key** for decryption. The keys are mathematically related but cannot be derived from each other easily, which ensures strong security.

**Key Points:**

1. **Two-Key System:**

   Each user has a **public key** (shared openly) and a **private key** (kept secret). Anyone can encrypt data using the public key, but only the owner can decrypt it using the private key.

2. **Secure Key Distribution:**

   Unlike symmetric encryption, there is **no need to share a secret key** beforehand. This eliminates the key distribution problem and supports secure communication over open networks.

3. **Confidentiality and Authentication:**

   Encryption with a public key ensures **confidentiality**.

   When data is encrypted or signed using a private key, it provides **authentication** and **non-repudiation** (digital signatures).

4. **Common Algorithms:**

   Public key encryption uses algorithms like **RSA**, **Elliptic Curve Cryptography (ECC)**, and **ElGamal**.

5. **Applications:**

   Used in **SSL/TLS for website security**, **email encryption**, **digital certificates**, **blockchain**, and **secure key exchange** in communication protocols.

**In short:**

Public key encryption uses a **public–private key pair** to provide **secure communication, authentication, and digital signatures**, making it suitable for use in modern internet and cybersecurity applications.

# Unit II

## 2. Authentication of Electronic records & E-Governance

**2.1 Authentication of Electronic records**
**2.2 Digital Signature**
**2.3 RSA Digital Signature**
**2.4 Hash Function**
**2.5 Working of Digital Signature**
**2.6 Electronic Governance**

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## 2.1 Authentication of Electronic records:

Authentication of electronic records refers to the process of **verifying the identity of the sender** and ensuring that the electronic document is **genuine, unaltered, and trustworthy**. Under the **Information Technology (IT) Act, 2000**, electronic records gain legal validity when properly authenticated.

**Key Points:**

1. **Use of Digital Signatures:**
   The IT Act recognizes **digital signatures** as the primary method to authenticate electronic records. A digital signature is created using the sender's **private key** and can be verified using the corresponding **public key**, ensuring authenticity.

2. **Ensures Integrity of Data:**
   Authentication confirms that the electronic record has **not been altered or tampered with** during transmission. Even a small change will invalidate the digital signature.

3. **Legal Recognition:**
   Section 3 of the IT Act legally recognizes digital signatures, making digitally signed electronic records valid and acceptable in courts, government processes, and business transactions.

4. **Use of Certifying Authorities (CAs):**
   Digital signatures are issued by licensed **Certifying Authorities** who verify the identity of the user. This adds trust to the authentication process.

5. **Applications:**
   Authentication of electronic records is used in **e-government services, e-commerce, online banking, tax filing, digital contracts**, and various secure communication systems.

**In short:**

Authentication of electronic records ensures that the document is **genuine, secure, legally valid, and unaltered**, mainly through the use of **digital signatures and public-key cryptography**.

## 2.2 Digital Signature:

A **digital signature** is an electronic method used to **authenticate the identity** of the sender and to ensure the **integrity** of an electronic document. It is created using **public key cryptography**, mainly through the use of a **private key** and verified using a **public key**. The **Information Technology (IT) Act, 2000** gives legal recognition to digital signatures in India.

**Key Points:**

1. **Authentication:**
   A digital signature verifies the identity of the sender. Since it is created using the sender's private key, only the owner can generate it.

2. **Integrity:**
   Digital signatures ensure that the content of an electronic record has **not been altered or tampered with** after signing. Any modification invalidates the signature.

3. **Non-Repudiation:**
   The sender cannot deny having signed the document because the signature is uniquely linked with their private key.

4. **How It Works:**
   The sender's computer generates a **hash** of the message.
   This hash is encrypted using the sender's **private key** to create the digital signature.
   The receiver uses the sender's **public key** to verify the signature.

5. **Legal and Practical Use:**
   Digital signatures are widely used in **e-filing of taxes**, **online banking**, **e-commerce**, **government services**, **signing PDF documents**, and **secure email communication**.
   Certifying Authorities (CAs) issue Digital Signature Certificates (DSCs) to users.

**In short:**

A digital signature is a **secure, legally recognized electronic signature** that provides **authentication, integrity, and non-repudiation** using public-key cryptography.

## 2.3 RSA Digital Signature:

The **RSA Digital Signature** is a technique that uses the **RSA public-key algorithm** to authenticate electronic documents. It ensures **authenticity, integrity, and non-repudiation** by using a pair of mathematically related keys: a **private key** for signing and a **public key** for verification.

**Key Points:**

1. **Uses Public-Key Cryptography:**
   RSA digital signatures are created using the sender's **private key**. The corresponding **public key** is used by the receiver to verify the signature. This ensures that only the genuine owner could have signed the document.

2. **Process of Creating a Digital Signature:**
   A **hash** (message digest) of the original message is generated using a hash function like SHA-256.
   This hash is then **encrypted with the sender's private key** to create the RSA digital signature.
   The signature is sent along with the message.

3. **Verification Process:**
   The receiver decrypts the signature using the sender's **public key** to recover the hash.
   The receiver also generates a fresh hash from the received message.
   If both hashes match, the signature is valid and the message is unaltered.

4. **Security Features:**

RSA signature provides:

**Authentication** (confirms the identity of the signer)

**Integrity** (detects any modification of data)

**Non-repudiation** (signer cannot deny signing)

Security depends on large prime factorization, making RSA computationally strong.

5. **Applications:**

Widely used in **digital certificates (SSL/TLS)**, **software signing**, **e-government services**, **document signing (PDF, e-filing)**, and **secure email systems** such as PGP.

**In short:**

An RSA digital signature authenticates the sender and protects the integrity of the message by using a **private key to sign** and a **public key to verify**, making it a secure and legally recognized method of digital authentication.

## 2.4 Hash Function:

A **hash function** is a mathematical algorithm that converts input data of **any size** into a **fixed-length output**, called a **hash value** or **message digest**. It is widely used in **cryptography** to ensure **data integrity** and security.

**Key Points:**

1. **Fixed-Length Output:**

Regardless of the size of the input message, a hash function always produces an output of fixed length. For example, **SHA-256** produces a 256-bit hash.

2. **One-Way Function:**

Hash functions are **irreversible**. It is computationally infeasible to retrieve the original data from the hash value, ensuring security.

3. **Integrity Verification:**

Even a small change in the input data results in a completely different hash value. This property helps detect **data tampering**.

4. **Common Hash Algorithms:**

Popular hash functions include **MD5**, **SHA-1**, **SHA-256**, and **SHA-3**. Modern systems prefer SHA-256 or higher for better security.

5. **Applications:**

Hash functions are used in **digital signatures**, **password storage**, **message authentication**, **file integrity checking**, and **blockchain technology**.

**In short:**

A hash function ensures **data integrity and security** by converting data into a unique, fixed length digest that is fast to compute and difficult to reverse.

## 2.5 Working of Digital Signature:

The working of a digital signature explains how an electronic document is **signed and verified** to ensure **authentication, integrity, and non-repudiation**. It mainly uses **hash functions** and **public-key cryptography**.

**Steps Involved:**

1. **Creation of Message Digest:**
   The sender applies a **hash function** (e.g., SHA-256) to the original message. This generates a fixed-length **hash value (message digest)** representing the content.

2. **Signing the Message:**
   The sender encrypts the hash value using their **private key**. This encrypted hash forms the **digital signature**.

3. **Transmission:**
   The sender sends the **original message along with the digital signature** to the receiver.

4. **Verification by Receiver:**
   The receiver decrypts the digital signature using the sender's **public key** to obtain the original hash.
   The receiver also applies the same hash function to the received message to generate a new hash.

5. **Validation:**
   If both hash values match, the digital signature is **valid**, confirming:
   **Authentication** of the sender
   **Integrity** of the message
   **Non-repudiation** (sender cannot deny signing)

**In short:**

A digital signature works by **hashing the message, encrypting the hash with a private key, and verifying it using the public key**, ensuring secure and trustworthy electronic communication.

## 2.6 Electronic Governance:

**Electronic Governance (E-Governance)** refers to the use of **information and communication technologies (ICT)** by the government to deliver services, exchange information, and interact with citizens, businesses, and other government departments in an **efficient, transparent, and secure** manner.

**Key Points:**

1. **Use of ICT in Government Services:**
   E-Governance uses computers, internet, and digital platforms to provide government services such as online applications, payments, certificates, and information access.

2. **Improved Efficiency and Transparency:**
   It reduces paperwork, delays, and corruption by automating processes and making government operations more transparent and accountable.

3. **Citizen-Centric Services:**

E-Governance enables citizens to access services anytime and anywhere, improving convenience and participation in governance.

4. **Legal Recognition under IT Act:**

The **Information Technology Act, 2000** provides legal validity to electronic records and digital signatures, supporting e-governance initiatives.

5. **Applications of E-Governance:**

Includes **e-filing of taxes**, **online voting systems**, **digital land records**, **Aadhaar services**, **e-procurement**, and **online grievance redressal systems**.

**In short:**

Electronic Governance uses **digital technology** to deliver government services efficiently, transparently, and securely, strengthening the relationship between the government and citizens.

# Unit III

## 3. Certifying Authorities

**3.1 Need of Certifying Authorities**
**3.2 Functioning of Certifying Authorities**
**3.3 Types of Certificates**
**3.4 Identification, Authorizing, Transactional certificate**
**3.5 Appointment of Controller**
**3.6 Functions**

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## 3.1 Need of Certifying Authorities:

**Certifying Authorities (CAs)** are trusted organizations authorized under the **Information Technology (IT) Act, 2000** to issue **Digital Signature Certificates (DSCs)**. They play a crucial role in establishing **trust, security, and legal validity** in electronic transactions.

**Key Points:**

1. **Authentication of Identity:**
   Certifying Authorities verify the identity of individuals, organizations, and government entities before issuing a Digital Signature Certificate. This ensures that the signer is genuine.

2. **Legal Validity of Digital Signatures:**
   Under the IT Act, digital signatures are legally valid only when issued by licensed Certifying Authorities, making electronic documents acceptable in courts and government processes.

3. **Trust in Electronic Transactions:**
   CAs act as a trusted third party, creating confidence among users in **e-commerce, e governance, and online banking** systems.

4. **Prevention of Fraud and Cybercrime:**
   By verifying identities and managing certificates, Certifying Authorities help prevent identity theft, impersonation, and unauthorized access.

5. **Management of Digital Certificates:**
   Certifying Authorities handle the **issuance, renewal, suspension, and revocation** of Digital Signature Certificates, ensuring secure and controlled use.

**In short:**

The need for Certifying Authorities arises to **verify identities, provide legal recognition, ensure trust, prevent fraud, and manage digital certificates**, making secure electronic communication possible.

## 3.2 Functioning of Certifying Authorities:

**Certifying Authorities (CAs)** are trusted entities licensed under the **Information Technology (IT) Act, 2000** to issue and manage **Digital Signature Certificates (DSCs)**. Their functioning ensures secure and legally valid electronic communication.

**Key Functions:**

1. **Identity Verification:**
   Certifying Authorities verify the identity of individuals, organizations, or government bodies by checking documents and credentials before issuing a Digital Signature Certificate.

2. **Issuance of Digital Signature Certificates:**
   After successful verification, the CA issues a DSC that links the subscriber's identity with their **public key**, enabling secure digital signing.

3. **Certificate Management:**
   CAs manage the entire life cycle of certificates, including **renewal, suspension, and revocation** in case of expiry, compromise, or misuse.

4. **Maintenance of Certificate Repository:**
   Certifying Authorities maintain an online database or repository of issued and revoked certificates so that users can verify the authenticity of a digital signature.

5. **Compliance with Legal and Security Standards:**
   CAs operate under the supervision of the **Controller of Certifying Authorities (CCA)** and follow prescribed security procedures to ensure trust and reliability.

**In short:**

The functioning of Certifying Authorities involves **verifying identities, issuing and managing digital certificates, maintaining repositories, and ensuring legal compliance**, which together build trust in electronic transactions.

## 3.3 Types of Certificates:

Digital certificates are electronic documents issued by **Certifying Authorities (CAs)** under the **Information Technology (IT) Act, 2000** to verify the identity of a person or organization in electronic transactions.

The main types of certificates are:

1. **Class 1 Certificate:**
   Issued for basic identity verification, mainly for **email communication**. It confirms the user's name and email address and is used for low-risk transactions.

2. **Class 2 Certificate:**
   Used for **moderate-risk transactions** such as **income tax e-filing, GST filing, company registration, and government portals**. The user's identity is verified against a trusted database.

3. **Class 3 Certificate:**
   Provides the **highest level of security**. Used for **e-tendering, e-procurement, e auction**, and other high-value online transactions. Physical presence or strong verification is required.

4. **Signature Certificate:**
   Used for **digitally signing documents** to ensure authenticity, integrity, and non repudiation. Commonly used for PDFs, legal documents, and government filings.

5. **Encryption Certificate:**

Used for **encrypting data and secure communication**. It ensures confidentiality of information exchanged over the internet.

**In short:**

Different types of digital certificates are issued based on the **level of security required**, ranging from basic identity verification to high-security transactions, ensuring trust and safety in electronic communication.

## 3.4 Identification, Authorizing, Transactional certificate:

### 1. Identification Certificate:

An **Identification Certificate** is a digital certificate used to **establish the identity** of an individual or organization in electronic systems. It confirms that a particular user is genuine and legally recognized.

**Key Points:**

1. It links a user's **identity details** (name, email ID, organization) with their **public key**.
2. Issued by a licensed **Certifying Authority (CA)** after identity verification.
3. Used for **login authentication** and identity proof in online systems.
4. Helps prevent **impersonation and identity theft**.
5. Commonly used in **e-governance portals, corporate networks, and secure email systems**.

**In short:**

Identification certificates are used to **prove who the user is** in electronic communication.

### 2. Authorizing Certificate:

An **Authorizing Certificate** is used to **grant permission or authority** to a user to perform specific actions in an electronic system.

**Key Points:**

1. It defines the **role and privileges** of the certificate holder.
2. Ensures that only **authorized users** can access or approve sensitive operations.
3. Used in systems requiring **role-based access control**.
4. Prevents **unauthorized access and misuse** of digital resources.
5. Commonly used in **banking systems, government approvals, e-procurement, and corporate workflows**.

**In short:**

Authorizing certificates control **what a user is allowed to do** in an electronic system.

### 3. Transactional Certificate:

A **Transactional Certificate** is used to **secure and authenticate electronic transactions** performed over digital networks.

**Key Points:**

1. Ensures **security, integrity, and authenticity** of online transactions.
2. Used to digitally sign or encrypt **transactional data**.
3. Protects financial and commercial transactions from **tampering and fraud**.
4. Provides **non-repudiation**, so parties cannot deny a transaction.

5. Widely used in **online banking, e-commerce, e-tendering, and payment gateways**.

**In short:**

Transactional certificates ensure that **electronic transactions are secure, trusted, and legally valid**.

## 3.5 Appointment of Controller:

The **Appointment of the Controller of Certifying Authorities (CCA)** is provided under the **Information Technology (IT) Act, 2000** to regulate and supervise the functioning of Certifying Authorities in India. The Controller ensures trust and security in digital signature systems.

**Key Points:**

1. **Authority for Appointment:**
   The **Central Government** appoints the Controller of Certifying Authorities under the provisions of the IT Act, 2000.

2. **Qualification and Experience:**
   The person appointed as Controller must possess **prescribed qualifications, experience, and expertise** in information technology, law, or cyber security.

3. **Powers and Responsibilities:**
   The Controller has powers to **license, regulate, and monitor Certifying Authorities** and ensure compliance with the Act and its rules.

4. **Assistance to Controller:**
   The Central Government may appoint **Deputy Controllers and Assistant Controllers** to help the Controller in carrying out duties effectively.

5. **Purpose of Appointment:**
   The appointment ensures **secure implementation of digital signatures**, proper functioning of Certifying Authorities, and **trust in electronic transactions and e governance**.

**In short:**

The Controller of Certifying Authorities is appointed by the **Central Government** to **regulate and supervise digital signature certification systems** under the IT Act.

## 3.6 Functions:

**Functions of the Controller of Certifying Authorities:**

The **Controller of Certifying Authorities (CCA)** is responsible for regulating and supervising the digital signature infrastructure in India.

The main functions are:

1. **Licensing of Certifying Authorities:**
   The Controller grants, renews, suspends, or revokes licenses of Certifying Authorities (CAs) after ensuring they meet legal and security requirements.

2. **Regulation and Supervision:**
   The Controller monitors the working of Certifying Authorities to ensure compliance with the IT Act, rules, and prescribed security standards.

3. **Specification of Standards:**
   The Controller specifies standards for **digital signatures, encryption methods, and security procedures** to be followed by Certifying Authorities.

4. **Maintenance of Public Key Infrastructure (PKI):**
   The Controller oversees the functioning of the **Public Key Infrastructure**, ensuring secure issuance and verification of Digital Signature Certificates.

5. **Investigation and Corrective Action:**
   The Controller can investigate complaints against Certifying Authorities and take corrective actions to prevent misuse or security breaches.

**In short:**

The functions of the Controller include **licensing, regulating, standard-setting, supervising Certifying Authorities, and ensuring secure digital signature systems**.

# Unit IV

## 4. Domain name Disputes and Cyber Crimes

**4.1 Background of Domain Names**

**4.2 Where lies the dispute?**

**4.3 Insertion of Internet Domain Names and the trademark Law**

**4.4 Classification of Cyber Crime, Target of computer crime**

**4.5 Damage to computer System: Unauthorized Access, Packet Sniffing, Tempest attack, Password Cracking, Butter overflow**

**4.6 Computer virus: Viruses, Logic Bomb, Worms, Trojan Horse Programme, Denial of Service, Tampering with Computer Source Documents**

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## 4.1 Background of Domain Names:

Domain names were introduced to make the **Internet user-friendly** by replacing complex numerical Internet Protocol (IP) addresses with easy-to-remember names. In the early days of the Internet, computers communicated only using IP addresses, which were difficult for humans to recall.

**Key Points:**

1. **Need for Domain Names:**

   Every device on the Internet has a unique **IP address** (e.g., 192.168.1.1). Domain names were created to map these numbers to meaningful names like **[www.example.com](http://www.example.com)**.

2. **Domain Name System (DNS):**

   The **Domain Name System (DNS)** was developed as a hierarchical and distributed naming system that translates domain names into corresponding IP addresses.

3. **Evolution of Domain Structure:**

   Domain names are structured in levels, such as **Top-Level Domains (TLDs)** (.com, .org, edu), **Country Code TLDs** (.in, .uk), and **Second-Level Domains** (example).

4. **Global Management:**

   The Internet Corporation for Assigned Names and Numbers (**ICANN**) was established to manage domain name allocation and ensure global uniqueness.

5. **Legal and Commercial Importance:**

   With the growth of **e-commerce and online services**, domain names gained commercial value, leading to legal issues like **trademark conflicts and cybersquatting**.

**In short:**

Domain names were developed to simplify Internet access by converting numeric IP addresses into meaningful names, forming a crucial part of Internet communication and online identity.

## 4.2 Where lies the dispute?

The dispute in domain names mainly lies in the **conflict between domain name registration and trademark rights**. Since domain names are unique and registered on a ‘first-come, first-served basis’, legal issues arise when they clash with existing brand names or trademarks.

**Key Points:**

1. **Domain Name vs Trademark Conflict:**
   A dispute arises when a domain name is identical or confusingly similar to a registered trademark, causing confusion among users.

2. **Cybersquatting:**
   Disputes often occur when a person registers a well-known trademark as a domain name with the intention of selling it to the rightful owner at a high price.

3. **Bad Faith Registration:**
   If a domain name is registered to mislead customers, damage reputation, or divert traffic for commercial gain, it leads to legal disputes.

4. **Jurisdictional Issues:**
   Domain names operate globally, while trademark laws are territorial, creating conflicts over which country's laws apply.

5. **Consumer Confusion and Brand Dilution:**
   Disputes arise when users are misled into believing a domain belongs to a genuine brand, harming business goodwill and consumer trust.

**In short:**

The dispute lies in the **unauthorized or bad-faith use of domain names that infringe trademark rights**, leading to confusion, misuse, and legal conflict.

## 4.3 Insertion of Internet Domain Names and the trademark Law:

### 1. Insertion of Internet Domain Names:

The insertion of Internet domain names refers to the **registration and inclusion of domain names in the Domain Name System (DNS)**, which gives an entity an online identity. Domain names are registered on a **first-come, first-served basis**, without initial verification of trademark rights.

**Key Points:**

1. Domain names are inserted into the **DNS database** through accredited registrars.
2. Registration does not automatically check for **existing trademarks**, leading to conflicts.
3. A registered domain name provides **global visibility and online identity**.
4. Issues arise when domain names are similar to **well-known brand names**.
5. This system has led to disputes such as **cybersquatting and passing off**.

**In short:**

Insertion of domain names creates online identity but can result in legal disputes when it conflicts with trademark rights.

### 2. Trademark Law:

**Trademark law** protects symbols, names, logos, and brand identities that distinguish goods or services of one entity from another. In India, trademarks are governed by the **Trade Marks Act, 1999**.

**Key Points:**

1. Trademarks protect **brand identity and goodwill** of businesses.
2. Rights are generally **territorial**, unlike domain names which are global.

3. Trademark law prevents **unauthorized use** of identical or confusingly similar marks.
4. Domain names are treated as **business identifiers**, similar to trademarks.
5. Trademark owners can take legal action against **infringing domain names**.

**In short:**

Trademark law safeguards brand identity and is used to resolve disputes where domain names infringe registered trademarks.

## 4.4 Classification of Cyber Crime, Target of computer crime:

### 1. Classification of Cyber Crime:

Cyber crimes are unlawful acts committed using computers, networks, or the internet. They can be classified based on the **target or victim** involved.

**Types of Cyber Crime:**

1. **Crimes Against Individuals:**
   Includes cyber stalking, online harassment, identity theft, phishing, email spoofing, and defamation.

2. **Crimes Against Property:**
   Involves damage to digital property such as data theft, hacking, malware attacks, software piracy, and intellectual property theft.

3. **Crimes Against Organizations:**
   Includes unauthorized access to corporate systems, data breaches, denial of service (DoS) attacks, and industrial espionage.

4. **Crimes Against Society:**
   Involves spreading obscene content, fake news, hate speech, online gambling, and circulation of illegal materials.

5. **Crimes Against Government:**
   Includes cyber terrorism, hacking government websites, espionage, and attacks on critical infrastructure.

**In short:**

Cyber crimes are classified based on the **victim affected**, ranging from individuals to governments.

### 2. Target of Computer Crime:

The target of computer crime refers to the **entities or resources attacked** by cyber criminals.

**Main Targets:**

1. **Individuals:**
   Personal data, bank details, passwords, and social media accounts are common targets.

2. **Businesses and Organizations:**
   Corporate data, customer databases, trade secrets, and financial systems are targeted for financial gain.

3. **Government Systems:**
Government databases, defense networks, and public service systems are targeted for political or terrorist motives.

4. **Computer Systems and Networks:**
Attackers target servers, websites, and networks to disrupt services or steal information.

5. **Digital Assets and Information:**
Includes intellectual property, software, confidential data, and digital currency.

**In short:**
Computer crimes target **people, data, systems, organizations, and governments**, aiming to steal, damage, or disrupt digital resources.

## 4.5 Damage to computer System: Unauthorized Access, Packet Sniffing, Tempest attack, Password Cracking, Butter overflow:

### 1. Unauthorized Access:

**Unauthorized access** refers to accessing a computer system, network, or data **without permission** of the owner. It is a serious cyber offence under cyber laws.

**Key Points:**
1. Occurs when an attacker bypasses security controls like login credentials or firewalls.
2. Can be done by hacking passwords, exploiting software vulnerabilities, or using stolen credentials.
3. Leads to **data theft, data modification, or system damage**.
4. Violates privacy and confidentiality of users.
5. Punishable under the **Information Technology Act, 2000**.

**In short:**
Unauthorized access damages computer systems by allowing illegal entry and misuse of data.

### 2. Packet Sniffing:

**Packet sniffing** is a technique where an attacker secretly **captures and monitors data packets** traveling over a network.

**Key Points:**
1. Attackers use sniffing tools to intercept network traffic.
2. Sensitive data like usernames, passwords, and bank details can be stolen.
3. Common in unsecured or public Wi-Fi networks.
4. Can lead to identity theft and financial fraud.
5. Prevented using encryption protocols like **HTTPS and VPNs**.

**In short:**
Packet sniffing damages computer systems by **stealing confidential data during transmission**.

**3. TEMPEST Attack:**

A **TEMPEST attack** is a sophisticated cyber attack where attackers capture **electromagnetic signals** emitted by electronic devices.

**Key Points:**

1. Computers emit electromagnetic radiation during operation.
2. Attackers intercept these signals to reconstruct data.
3. No physical access to the system is required.
4. Used mainly in **espionage and military cyber attacks**.
5. Prevented using shielding and secure hardware.

**In short:**

TEMPEST attacks compromise systems by leaking data through electromagnetic emissions.

**4. Password Cracking:**

**Password cracking** is the process of illegally obtaining passwords to gain unauthorized access to systems.

**Key Points:**

1. Uses methods like brute force, dictionary attacks, and keylogging.
2. Weak or simple passwords are easy targets.
3. Enables attackers to access private accounts and data.
4. Can lead to identity theft and system misuse.
5. Prevented by strong passwords and two-factor authentication.

**In short:**

Password cracking damages computer systems by breaking authentication security.

**5. Buffer Overflow (Butter Overflow):**

**Buffer overflow** is a programming error where more data is written into memory than it can handle.

**Key Points:**

1. Occurs due to poor input validation in software.
2. Excess data overwrites adjacent memory locations.
3. Attackers can execute malicious code.
4. Can cause system crashes or unauthorized control.
5. Prevented by secure coding practices.

**In short:**

Buffer overflow attacks damage computer systems by exploiting software weaknesses to gain control.

## 4.6 Computer virus: Viruses, Logic Bomb, Worms, Trojan Horse Programme, Denial of Service, Tampering with Computer Source Documents:

**1. Computer Virus:**

A **computer virus** is a malicious program that attaches itself to legitimate files and spreads when the

infected file is executed.

**Key Points:**

1. Requires a **host program** to operate and spread.
2. Replicates itself and infects other files or systems.
3. Can corrupt data, slow down systems, or crash computers.
4. Spreads through email attachments, USB drives, and downloads.
5. Prevented using updated **antivirus software**.

**In short:**

A virus damages systems by infecting files and spreading malicious code.

**2. Logic Bomb:**

A **logic bomb** is malicious code that activates when a **specific condition or time** is met.

**Key Points:**

1. Remains dormant until triggered.
2. Trigger may be a date, event, or action.
3. Used to delete files or damage systems.
4. Often planted by insiders or disgruntled employees.
5. Difficult to detect before activation.

**In short:**

Logic bombs silently damage systems when preset conditions are fulfilled.

**3. Worms:**

A **worm** is a standalone malicious program that **replicates itself** without human intervention.

**Key Points:**

1. Does not require a host program.
2. Spreads automatically through networks.
3. Consumes system resources and network bandwidth.
4. Can slow down or crash servers.
5. Prevented by firewalls and security patches.

**In short:**

Worms damage networks by self-replication and resource exhaustion.

**4. Trojan Horse Programme:**

A **Trojan Horse** is a malicious program disguised as a legitimate or useful application.

**Key Points:**

1. Appears harmless but contains hidden malware.
2. Does not self-replicate like a virus.
3. Creates backdoors for attackers.
4. Used to steal data or gain unauthorized access.
5. Spread through fake software downloads and emails.

**In short:**

Trojan programs deceive users and allow attackers to control systems.

**5. Denial of Service (DoS) Attack:**

A **Denial of Service (DoS)** attack aims to make a computer or network **unavailable to users**.

**Key Points:**

1. Floods the system with excessive requests.
2. Overloads servers and crashes services.
3. Distributed DoS (DDoS) uses multiple systems.
4. Commonly targets websites and online services.
5. Causes financial and service disruption.

**In short:**

DoS attacks damage systems by interrupting normal services.

**6. Tampering with Computer Source Documents:**

Tampering with computer source documents refers to **unauthorized modification or destruction of source code**.

**Key Points:**

1. Includes altering, deleting, or hiding source code.
2. Affects software functionality and security.
3. Often done to introduce malicious code.
4. Considered a serious offence under **IT Act, 2000 (Section 65)**.
5. Can cause system failure or data loss.

**In short:**

Tampering with source documents damages computer systems and software integrity.